Customer Churn & Retention Analytics Report

Executive Summary

This project presents a comprehensive **Customer Churn and Retention Analytics** study using Python, SQLite3, and Power BI. The objective is to analyze customer behavior, identify major churn patterns, quantify recurring revenue exposure, and provide actionable recommendations for improving customer retention.

The analysis was performed on **5,021 customer records**, of which **1,540 customers were identified as churned** and **3,481 customers remained active**. The overall churn rate was **30.67%**, while the retention rate was **69.33%**.

The analysis identified the **Basic plan** as the segment with the highest churn rate at **31.81%**. Monthly-contract customers also showed greater churn instability compared with annual-contract customers. The estimated monthly recurring revenue associated with churned customers was **₹77,089**, equivalent to approximately **₹9.25 lakh on an annualized basis**.

The final Power BI dashboard converts these findings into interactive KPIs and visualizations, enabling users to analyze churn by plan type, contract type, geography, and other customer characteristics.

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1. Introduction

1.1 Background

Customer churn is one of the major challenges faced by subscription-based businesses. When customers discontinue their subscriptions, organizations lose recurring revenue and may also incur additional costs to acquire replacement customers.

Traditional reporting methods often show only the number of customers who have already churned. However, analyzing customer characteristics and behavioral patterns can help organizations understand **which customer segments experience higher churn and where retention efforts should be focused**.

This project applies data analytics techniques to customer, subscription, and support data to identify churn patterns and quantify their potential business impact.

1.2 Problem Statement

Organizations often face difficulty in identifying customer segments associated with high churn. Without proper analytical visibility, retention activities may become reactive rather than data-driven.

The major challenges addressed by this project are:

- Difficulty in identifying customer segments with higher churn rates
- Limited visibility into churn patterns across plans and contracts
- Difficulty in quantifying the recurring revenue associated with churned customers
- Lack of an interactive dashboard for monitoring churn-related KPIs
- Difficulty in converting raw customer data into actionable business insights

1.3 Project Objectives

The primary objectives of this project are:

- Calculate customer churn and retention metrics
- Analyze churn patterns across different customer segments
- Compare churn rates across plan and contract types
- Analyze the relationship between customer churn and revenue exposure
- Develop an interactive Power BI dashboard
- Create reusable DAX measures for churn-related KPIs
- Provide data-driven recommendations for customer retention

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2. Technology Stack and Methodology

2.1 Technology Stack

Component	Technology	Purpose
Database	SQLite3	Data storage and SQL-based data retrieval
Programming	Python	Data processing and analysis

Data Analysis	Pandas, NumPy	Data cleaning, transformation and feature engineering
Visualization	Matplotlib, Seaborn	Exploratory data analysis
Business Intelligence	Power BI	Interactive dashboard development
Calculations	DAX	KPI and analytical measure creation

2.2 Data Sources

The project uses three SQLite database tables:

1. Customers Table

Contains customer-level information such as:

- Customer ID
- Gender
- State
- Demographic attributes

2. Subscriptions Table

Contains subscription-related information such as:

- Plan type
- Contract type
- Monthly charges
- Subscription start date
- Cancellation date

3. Support Table

Contains customer service interaction information such as:

- Support interactions
- Customer issues
- Service-related information

These datasets were combined to create a unified analytical dataset.

2.3 Data Processing Pipeline

The overall analytical workflow is:

SQLite3 Database
↓
Data Extraction using Python
↓
Data Cleaning & Transformation
↓
Feature Engineering
↓
Exploratory Data Analysis
↓
Power BI Data Model
↓
DAX Measures
↓
Interactive Dashboard
↓
Business Insights & Recommendations

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3. Data Preprocessing and Feature Engineering

3.1 Data Preparation

The following preprocessing activities were performed:

1. Extracted data from SQLite3 using Python.
2. Loaded the tables using Pandas.
3. Combined relevant tables using customer identifiers.
4. Standardized column names.
5. Removed irrelevant or inconsistent records.
6. Standardized categorical values.
7. Converted date columns into appropriate date formats.
8. Created churn-related features.
9. Prepared the final dataset for analysis and visualization.

3.2 Churn Flag Creation

A binary `churn_flag` was created based on the customer's cancellation status.

The interpretation is:

- `1` = Customer has churned
- `0` = Customer is currently active

Conceptually:

If cancellation_date is available → churn_flag = 1
If cancellation_date is NULL → churn_flag = 0

This binary field was used for calculating churn-related KPIs in Power BI.

3.3 Final Dataset

The final analytical dataset contains:

- **Records:** 5,021
- **Columns:** 21
- **Primary analytical fields:** plan_type, contract_type, monthly_charges, cancellation_date, churn_flag

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4. Exploratory Data Analysis

4.1 Overall Churn Metrics

Metric	Value
Total Customers	5,021
Churned Customers	1,540
Active Customers	3,481
Churn Rate	**30.67%**

Retention Rate	**69.33%**

The results show that approximately **31 out of every 100 customers have churned**, while approximately **69 out of every 100 customers remain active**.

4.2 Plan-Wise Churn Analysis

Plan Type	Churn Rate	Risk Level
Basic	**31.81%**	Highest
Standard	**30.48%**	Medium
Premium	**29.77%**	Lowest

Key Finding

The **Basic plan has the highest churn rate at 31.81%**, followed by the Standard plan at 30.48%. The Premium plan has the lowest churn rate among the three segments at 29.77%.

Therefore, the Basic-plan segment should receive particular attention when designing retention strategies.

4.3 Contract-Wise Churn Analysis

Contract Type	Observed Pattern
Monthly	Higher churn tendency
Annual	More stable customer behavior

Key Finding

Monthly-contract customers demonstrate a higher tendency to churn compared with annual-contract customers. This suggests that longer contractual commitments may be associated with greater customer stability.

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5. Revenue Impact Analysis

5.1 Revenue at Risk

The project estimates the monthly recurring revenue associated with churned customers as:

Monthly Revenue at Risk = ₹77,089

The approximate annualized revenue exposure is:

₹77,089 × 12 = ₹9,25,068

Therefore:

Annualized Revenue Exposure ≈ ₹9.25 lakh

This metric helps translate customer churn into a financial impact that can be understood by business stakeholders.

5.2 Revenue Exposure by Customer Segment

Revenue-at-risk analysis can be used to identify customer segments where churn has the greatest financial impact.

While churn rate identifies **where customers are leaving at a higher rate**, revenue-at-risk analysis identifies **where customer loss has greater financial consequences**.

This distinction is important because a segment with a lower churn rate may still generate significant revenue exposure if its customers have higher monthly charges.

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6. Power BI Dashboard

6.1 Dashboard Overview

An interactive Power BI dashboard was developed to provide a centralized view of customer churn and retention performance.

The dashboard includes KPI cards, filters, and visualizations that allow users to analyze customer churn from multiple perspectives.

6.2 KPI Cards

The primary KPIs include:

1. Total Customers
2. Churned Customers
3. Churn Rate
4. Retention Rate
5. Revenue at Risk
6. Average Churn Score, where applicable

6.3 Dashboard Filters

The dashboard provides interactive filtering based on relevant customer attributes, including:

- Plan Type
- Contract Type
- Gender
- State

6.4 Dashboard Visualizations

The dashboard includes analytical visualizations such as:

- Churn Rate by Plan Type
- Churn Rate by Contract Type
- Revenue at Risk
- Geographic Churn Analysis
- Customer Churn Distribution

These visualizations allow users to identify high-risk segments and compare customer behavior interactively.

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7. DAX Measures

7.1 Total Customers

Total Customers =
COUNTROWS(Sheet1)

7.2 Churned Customers

```
Churned Customers =
CALCULATE(
   COUNTROWS(Sheet1),
   Sheet1[churn_flag] = 1
)
```

7.3 Churn Rate

```
Churn Rate =
DIVIDE(
   [Churned Customers],
   [Total Customers],
   0
)
```

7.4 Retention Rate

```
Retention Rate =
DIVIDE(
   [Total Customers] - [Churned Customers],
   [Total Customers],
   0
)
```

7.5 Revenue at Risk

```
Revenue at Risk =
CALCULATE(
   SUM(Sheet1[monthly_charges]),
   Sheet1[churn_flag] = 1
)
```

This measure calculates the monthly recurring charges associated with customers identified as churned.

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8. Key Findings

8.1 Churn Patterns

The analysis produced the following major findings:

1. The overall customer churn rate is **30.67%**.
2. The retention rate is **69.33%**.
3. The Basic plan has the highest churn rate at **31.81%**.
4. The Premium plan has the lowest churn rate at **29.77%**.
5. Monthly contracts show greater churn instability compared with annual contracts.

8.2 Revenue Exposure

The analysis estimates:

- **₹77,089 monthly recurring revenue at risk**
- **Approximately ₹9.25 lakh annualized revenue exposure**

Revenue-at-risk analysis provides an additional financial perspective beyond simply counting churned customers.

8.3 Business Opportunity

The 69.33% retention rate indicates that the majority of customers remain active. Improving retention among the highest-risk segments could therefore help protect recurring revenue and improve overall customer lifetime value.

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9. Business Recommendations

9.1 Priority Actions

Priority	Recommended Action	Expected Benefit
1	Focus retention campaigns on Basic-plan and monthly-contract customers	Address segments with higher churn rates
2	Investigate price-related cancellation patterns	Identify potential price sensitivity

3	Monitor revenue at risk by customer segment	Prioritize financially important customers
4	Analyze support-related churn patterns	Identify service issues contributing to cancellations
5	Monitor churn KPIs regularly through Power BI	Enable continuous data-driven decision-making

9.2 Recommended Retention Strategy

Step 1: Identify High-Risk Segments

Use the dashboard to identify customer segments with higher churn rates.

Step 2: Prioritize Financial Impact

Combine churn information with monthly charges to identify customers or segments representing significant revenue exposure.

Step 3: Investigate Churn Reasons

Analyze cancellation and support information to understand whether price, service quality, contract structure, or other factors contribute to churn.

Step 4: Apply Targeted Retention Actions

Instead of applying the same strategy to all customers, design targeted retention campaigns for high-risk segments.

Step 5: Monitor Results

Use Power BI KPIs to continuously monitor churn and evaluate whether retention initiatives are improving customer outcomes.

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10. Conclusion and Future Scope

10.1 Project Outcomes

The project successfully demonstrates an end-to-end customer churn analytics workflow using Python, SQLite3, and Power BI.

The major outcomes include:

- Extracted and analyzed customer data from SQLite3
- Performed data cleaning and preprocessing using Python
- Created churn-related analytical features
- Conducted exploratory data analysis
- Developed reusable DAX measures
- Created an interactive Power BI dashboard
- Identified high-churn customer segments
- Quantified monthly revenue exposure
- Developed data-driven customer retention recommendations

The analysis identified a **30.67% churn rate**, a **69.33% retention rate**, and approximately **₹77,089 in monthly recurring revenue associated with churned customers**.

10.2 Future Scope

The project can be extended in several ways:

Initiative	Future Enhancement
Machine Learning	Develop predictive models to identify customers likely to churn before cancellation
Real-Time Analytics	Implement real-time monitoring of customer behavior and churn indicators
Automated Alerts	Generate alerts when high-risk customers are identified
Customer Lifetime Value	Combine churn probability with Customer Lifetime Value for better prioritization
CRM Integration	Integrate churn insights with CRM platforms to support automated retention workflows
Advanced Segmentation	Use clustering techniques to identify distinct customer behavior groups

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11. Appendices

Appendix A: Data Dictionary

Field	Description
`customer_id`	Unique identifier assigned to each customer
`plan_type`	Customer subscription plan such as Basic, Standard, or Premium
`contract_type`	Type of subscription contract
`monthly_charges`	Recurring monthly charge associated with the customer
`cancellation_date`	Date on which the customer cancelled the subscription
`churn_flag`	Binary indicator where 1 = churned and 0 = active
`churn_score`	Churn-related score/risk indicator, depending on the implemented dataset definition

Appendix B: Technical Requirements

Software

- Python 3.x
- SQLite3
- Power BI Desktop

Python Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn

Skills Used

- SQL
- Python
- Data Cleaning
- Exploratory Data Analysis
- Data Visualization
- Power BI

- DAX
- Business Analytics

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12. Project Information

Project Title: Customer Churn & Retention Analytics

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